



Benchmarking Spatial, Spectral, and Self-Supervised Cues for Face Forgery Detection under Realistic Degradation

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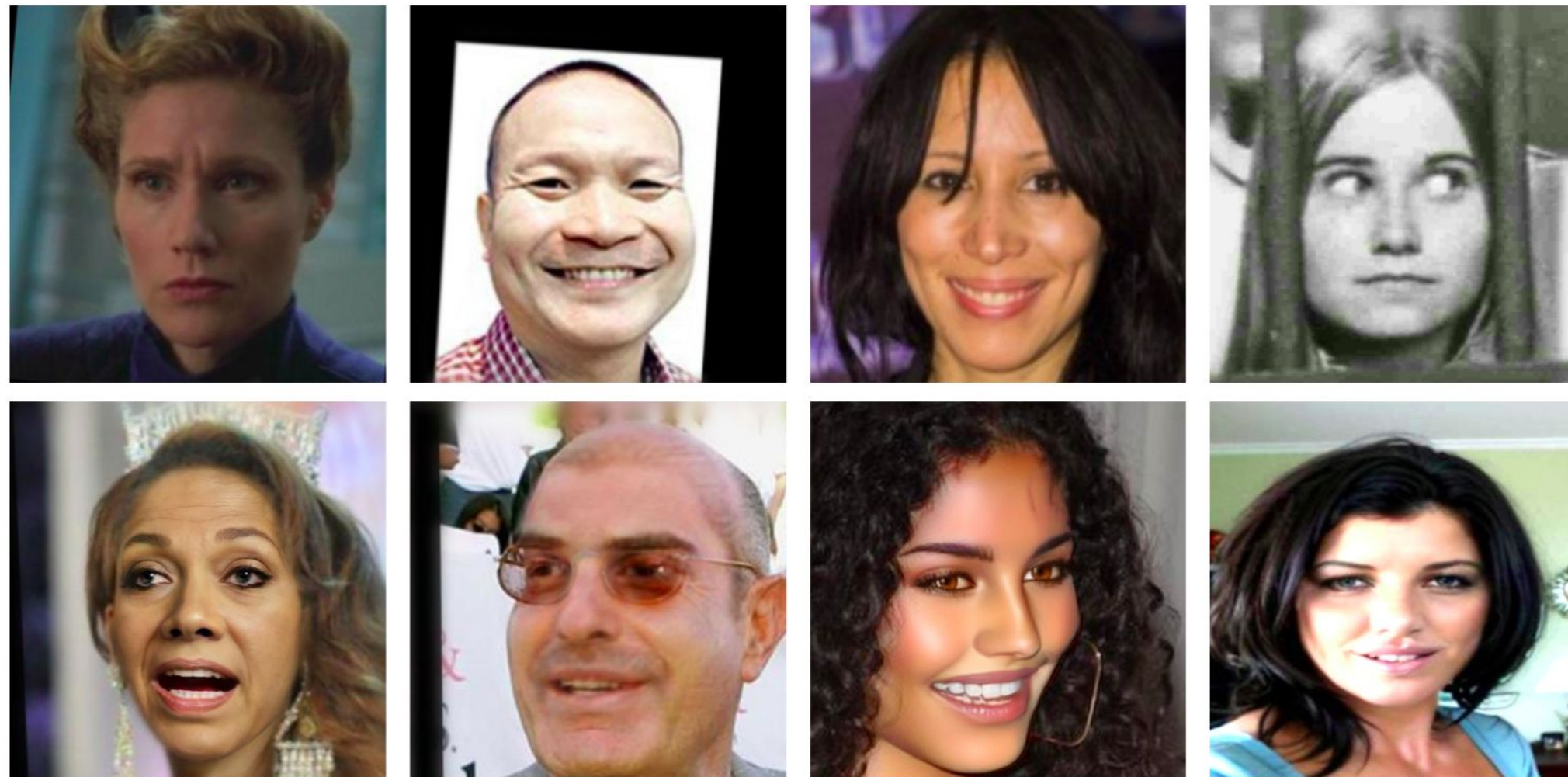
Introduction - Contextualization

- Easy access to realistic synthetic faces and identity manipulation tools
- Common detector benchmarks often **rely on controlled data**
- Images shared online are routinely compressed, resized, blurred, or re-encoded



Introduction - Contextualization

- The MFFI dataset [1] was recently introduced (2025) to study this gap
 - 50 forgery techniques, diverse authentic sources, multi-level degradations



Examples of **authentic (top)** and **forged (bottom)** face images from the MFFI





Introduction - Contributions

- Benchmark of **six heterogeneous architectures** on both **clean** and **degraded** MFFI partitions
- Quantification of the effect of **seven spatial, spectral, and hybrid input representations**
- Comparison of our results with state-of-the-art results on MFFI



Experimental Setup - Dataset

Dataset and Task

- Binary classification task (real vs. fake)
- ~57% fake
- Paired evaluation on **clean**

Test and **degraded Test-Hard**



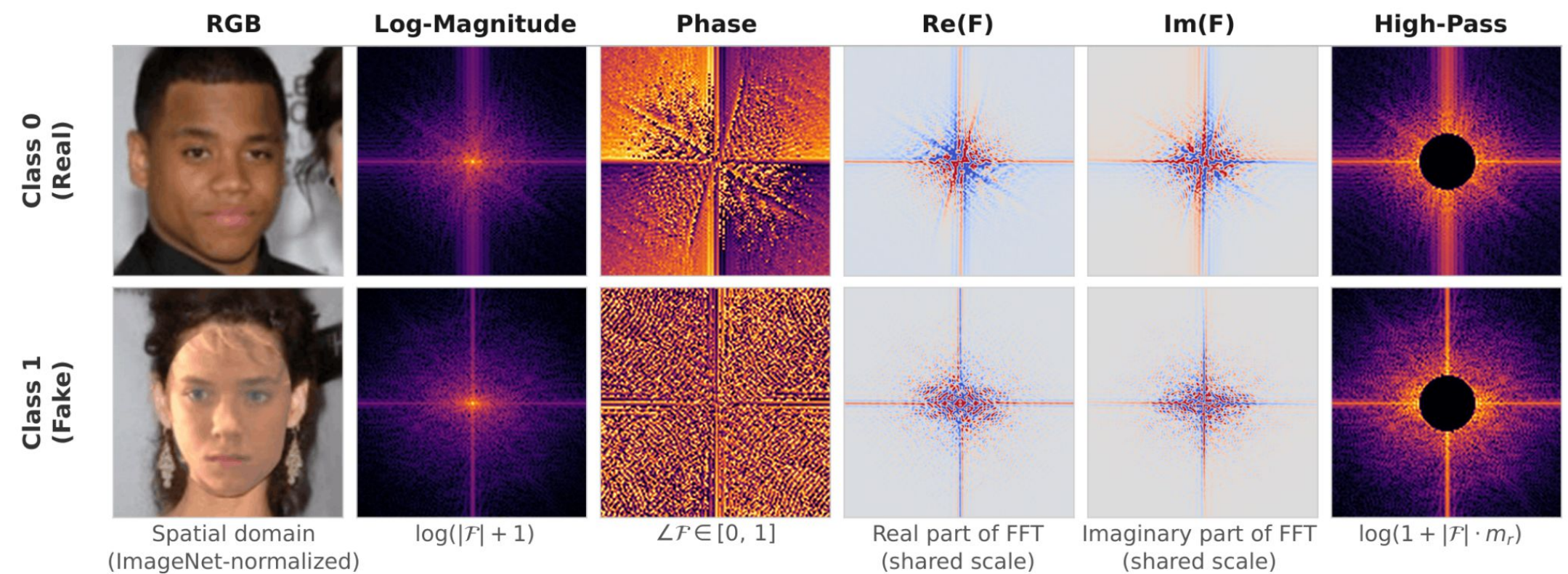
Examples of **Test (top)** and degraded **Test-Hard (bottom)** face images from the MFFI



Experimental Setup - Input Representations

Spatial

- RGB (3 ch)



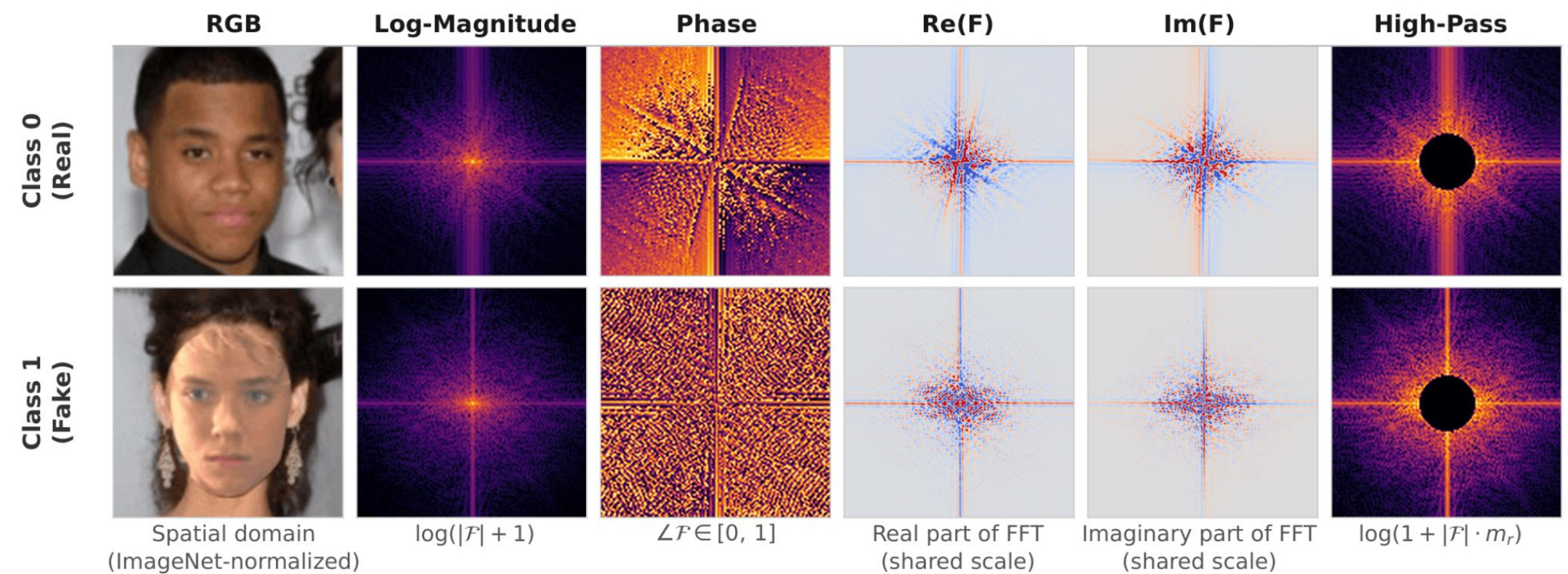
Spatial and frequency-domain views of authentic and forged face images.



Experimental Setup - Input Representations

Spectral

- Log-Magnitude (1 ch)
- Phase (1 ch)
- Complex Spectrum (2 ch)
- High-Pass Spectrum (1 ch)

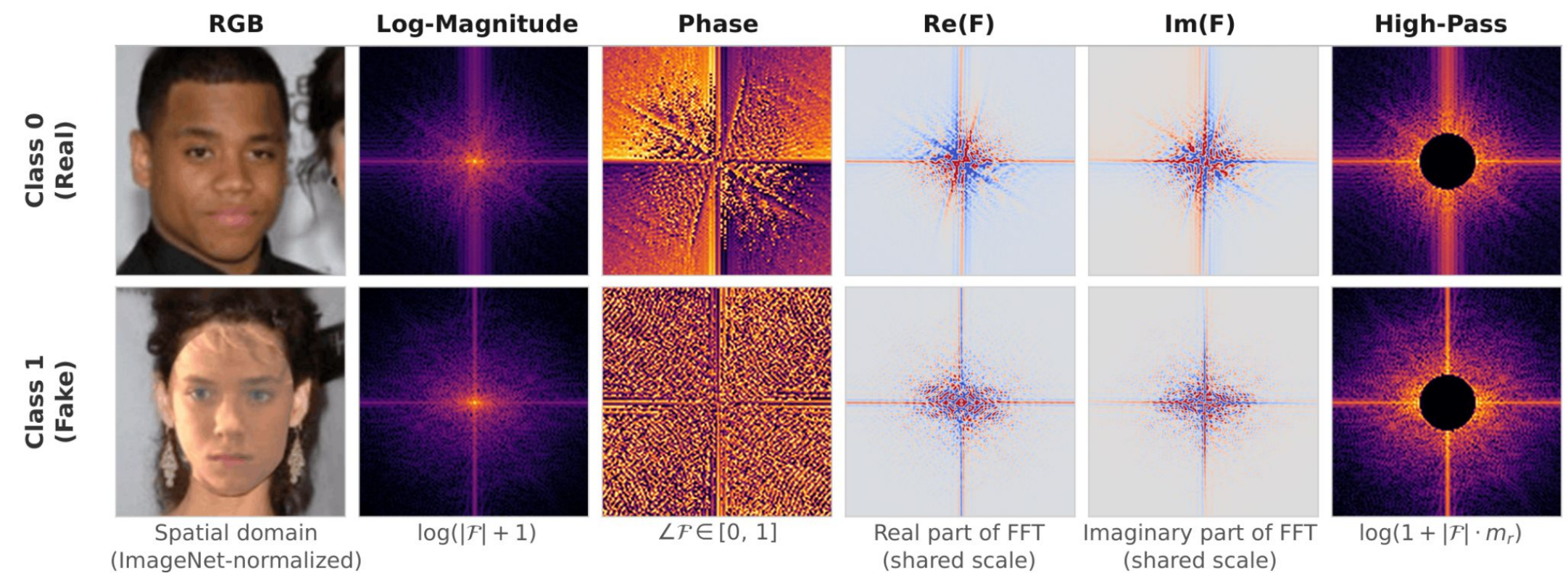


Spatial and frequency-domain views of authentic and forged face images.

Experimental Setup - Input Representations

Hybrid

- RGB+Mag (4 ch)
- RGB+Freq (6 ch)



Spatial and frequency-domain views of authentic and forged face images.



Experimental Setup - Architectures

- **Convolutional Models**

- Xception, ResNet-18, and MobileNetV3

- **Vision Transformer**

- ViT-B16 – 3 layers, 4 heads, 16x16 patch size

- **Language-Vision Encoder**

- CLIP-style visual encoder

- **Self-Supervised Backbone**

- DINOv3 frozen backbone with only a lightweight linear head trained.





Experimental Setup - Protocol & Metrics

1

Optimization Pipeline

AdamW, cross-entropy loss, weighted random sampler, and 3 runs.

2

Evaluation Metrics

ROC-AUC (threshold-independent) and **Accuracy** with the threshold fixed on the validation set.

3

Model Interpretability

Grad-CAM for CNNs and DINOv3; **Attention Rollout** for Transformers.



Results

- **Xception** is our best-performing model on the clean test (ROC-AUC **0.884**)
- Spectral channels alone **do not** outperform RGB

Model	Representation	Test (clean)		Test-Hard (degraded)	
		ROC-AUC	ACC	ROC-AUC	ACC
Xception [13]	<i>RGB</i>	0.884 ± 0.0091	0.757 ± 0.0051	0.609 ± 0.0196	0.568 ± 0.0033
	<i>Log-Magnitude</i>	0.686 ± 0.0125	0.638 ± 0.0090	0.517 ± 0.0200	0.535 ± 0.0292
	<i>Phase</i>	0.802 ± 0.0228	0.731 ± 0.0174	0.573 ± 0.0031	0.571 ± 0.0030
	<i>Complex Spectrum</i>	0.790 ± 0.0274	0.720 ± 0.0266	0.608 ± 0.0282	0.577 ± 0.0238
	<i>RGB+Mag</i>	0.878 ± 0.0152	0.768 ± 0.0115	0.650 ± 0.0231	0.603 ± 0.0189
	<i>High-Pass Spectrum</i>	0.674 ± 0.0196	0.622 ± 0.0116	0.501 ± 0.0174	0.520 ± 0.0132
	<i>RGB+Freq</i>	0.865 ± 0.0038	0.777 ± 0.0105	0.609 ± 0.0536	0.580 ± 0.0329
ResNet-18 [14]	<i>RGB</i>	0.846 ± 0.0201	0.725 ± 0.0238	0.635 ± 0.0001	0.573 ± 0.0001
	<i>Log-Magnitude</i>	0.703 ± 0.0289	0.647 ± 0.0210	0.533 ± 0.0003	0.547 ± 0.0002
	<i>Phase</i>	0.730 ± 0.0335	0.676 ± 0.0253	0.583 ± 0.0002	0.579 ± 0.0003
	<i>Complex Spectrum</i>	0.758 ± 0.0490	0.694 ± 0.0410	0.569 ± 0.0000	0.568 ± 0.0002
	<i>RGB+Mag</i>	0.841 ± 0.0226	0.742 ± 0.0211	0.634 ± 0.0003	0.576 ± 0.0000
	<i>High-Pass Spectrum</i>	0.685 ± 0.0135	0.633 ± 0.0150	0.519 ± 0.0000	0.538 ± 0.0000
	<i>RGB+Freq</i>	0.831 ± 0.0310	0.746 ± 0.0269	0.648 ± 0.0001	0.600 ± 0.0001
MobileNetV3 [12]	<i>RGB</i>	0.839 ± 0.0313	0.751 ± 0.0212	0.594 ± 0.0001	0.563 ± 0.0002
	<i>Log-Magnitude</i>	0.669 ± 0.0294	0.628 ± 0.0271	0.503 ± 0.0002	0.511 ± 0.0001
	<i>Phase</i>	0.689 ± 0.0252	0.646 ± 0.0215	0.561 ± 0.0002	0.572 ± 0.0000
	<i>Complex Spectrum</i>	0.733 ± 0.0443	0.671 ± 0.0361	0.543 ± 0.0002	0.560 ± 0.0000
	<i>RGB+Mag</i>	0.828 ± 0.0244	0.746 ± 0.0181	0.612 ± 0.0000	0.587 ± 0.0000
	<i>High-Pass Spectrum</i>	0.653 ± 0.0245	0.622 ± 0.0082	0.519 ± 0.0497	0.520 ± 0.0330
	<i>RGB+Freq</i>	0.798 ± 0.0231	0.726 ± 0.0195	0.548 ± 0.0499	0.539 ± 0.0330
ViT [15]	<i>RGB</i>	0.624 ± 0.0412	0.581 ± 0.0069	0.608 ± 0.0273	0.581 ± 0.0069
	<i>Log-Magnitude</i>	0.551 ± 0.0684	0.577 ± 0.0215	0.550 ± 0.0692	0.577 ± 0.0215
	<i>Phase</i>	0.576 ± 0.0343	0.578 ± 0.0088	0.567 ± 0.0278	0.577 ± 0.0088
	<i>Complex Spectrum</i>	0.608 ± 0.0181	0.574 ± 0.0002	0.601 ± 0.0093	0.574 ± 0.0002
	<i>RGB+Mag</i>	0.645 ± 0.0629	0.597 ± 0.0144	0.630 ± 0.0482	0.595 ± 0.0159
	<i>High-Pass Spectrum</i>	0.488 ± 0.0224	0.572 ± 0.0013	0.485 ± 0.0206	0.572 ± 0.0013
	<i>RGB+Freq</i>	0.633 ± 0.0025	0.596 ± 0.0074	0.582 ± 0.0477	0.581 ± 0.0147
DINOv3 [17]	<i>RGB</i>	0.809 ± 0.0046	0.712 ± 0.0158	0.726 ± 0.0000	0.663 ± 0.0022
CLIP-style enc. [16]	<i>RGB</i>	0.750 ± 0.0043	0.683 ± 0.0015	0.619 ± 0.0006	0.581 ± 0.0008

Results - Clean vs. Degraded Performance

Performance under RGB Input (Mean ROC-AUC)

Model Family	Test (Clean) ROC-AUC	Test-Hard (Degraded) ROC-AUC	Δ (Drop)
Xception	0.884	0.609	-27.5 p.p.
ResNet-18	0.846	0.635	-21.1 p.p.
MobileNetV3	0.839	0.594	-24.5 p.p.
ViT	0.624	0.608	-1.6 p.p.
CLIP-style enc.	0.750	0.619	-13.1 p.p.
DINOv3 (frozen)	0.809	0.726	-8.3 p.p.

Key finding: Clean performance does not predict robustness under compression, resizing, and blur.





Results - Input Representation Analysis

On Test (Clean)

- Spatial RGB consistently outperformed Log-Magnitude, Phase, and High-Pass **alone**
- RGB+Mag (4 ch) and RGB+Freq (6 ch) outperform pure RGB **under degradation for CNNs**
- Fourier features serve as valuable auxiliary evidence, **not standalone replacements**

On Test-Hard (Degraded)

- DINOv3 outperformed the other models under degradation even **when compared with frequency cues**



Results - Comparison with State of the Art

Image-Level ROC-AUC on MFFI Benchmark

Method	Training Data	Test (Clean)	Test-Hard (Degraded)
Xception baseline [1]	MFFI	0.852	0.708
SRM [1]	MFFI	0.877	0.580
RFM [1]	MFFI	0.849	0.720
Ours (Xception, RGB)	MFFI	0.884	0.609
Ours (DINOv3, frozen)	MFFI	0.809	0.726
<i>DeFakerOne [2]</i>	<i>12.5M multi-domain, including MFFI</i>	<i>0.961</i>	<i>n/a</i>

Within comparable MFFI training protocols, our benchmark matches or exceeds published baselines.



Qualitative Evidence - Attribution Maps (RGB)

- 1 **Convolutional Detectors**
Concentrate on localized artifacts in forged images
- 2 **Transformers**
Produce diffuse, scattered attention patterns
- 3 **Self-Supervised DINOv3**
Distributes attention across broad facial landmarks.

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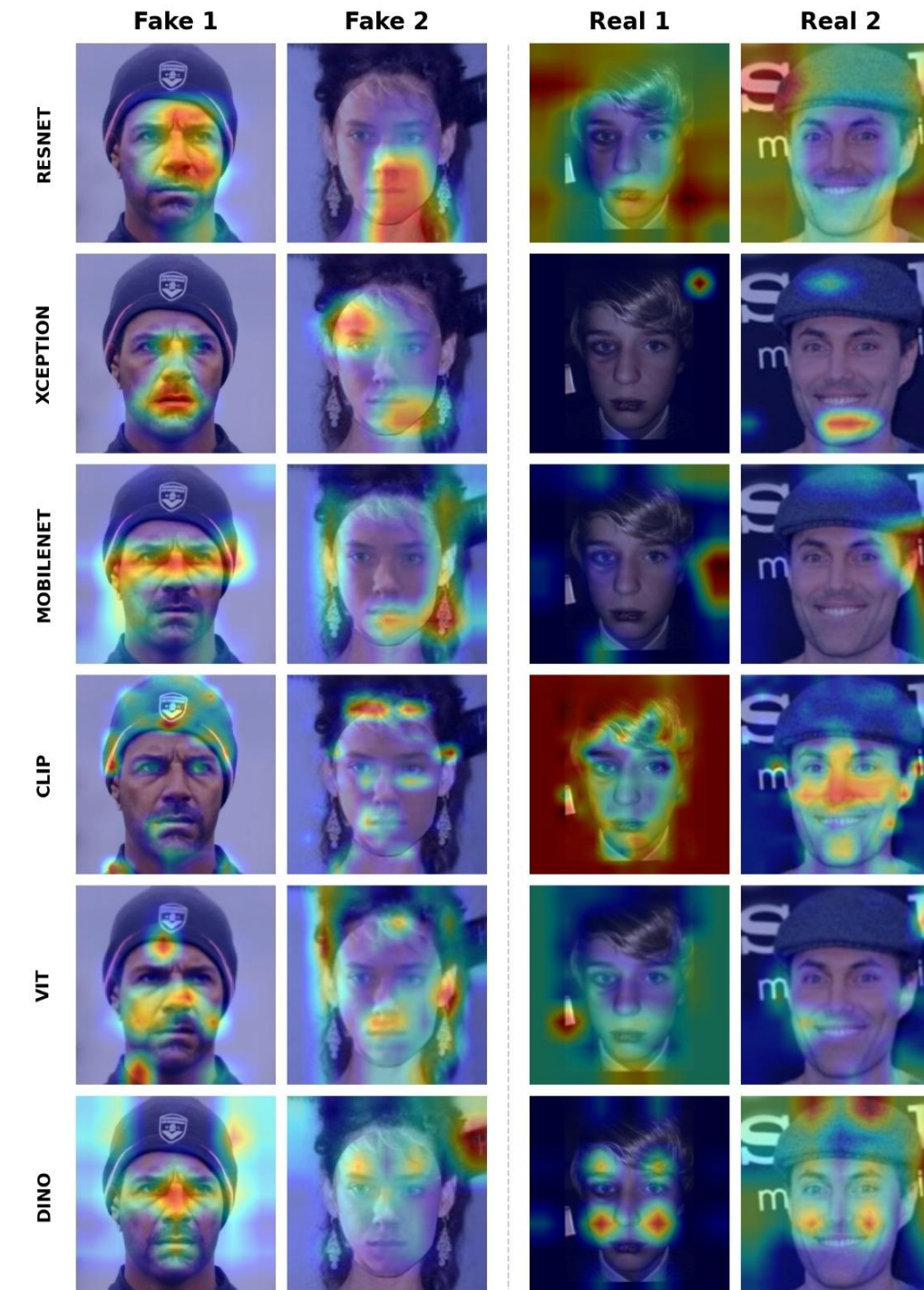


Image matrix comparing explainability heatmaps (Grad-CAM and Attention Rollout) for six models across two forged faces and two real faces.



Discussion & Limitations

- Measures robustness to MFFI degradations rather than multi-dataset generalization.
- Five models trained from scratch vs. one large-scale self-supervised pretrained backbone.
- DINOv3 evaluated only with a linear probe; fine-tuning effects remain to be explored.





Conclusions & Future Work

- Evaluating detectors under realistic perturbations is essential for reliable deepfake forensics.
- DINOv3 provides superior degradation robustness (0.726 ROC-AUC) with minimal training overhead.
- **Future Directions:**
 - Cross-dataset evaluation, fine-tuning backbones, and diffusion-specific attacks.



References

- **[1]** MIAO, Changtao et al. (2025). MFFI: Multi-dimensional face forgery image dataset for real-world scenarios. Proceedings of the 33rd ACM International Conference on Multimedia. p. 13235-13242.
- **[2]** GuangJian Team. (2026). Venus-DeFakerOne: Unified fake image detection & localization. arXiv preprint arXiv:2605.14091



GitHub



Thank you!

Questions?

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